



Choose the correct answer.

**Consider the circuit shown in Figure 1, where the current source has $I_s = 2\angle 0^\circ$ A rms, and answer questions 1-4.

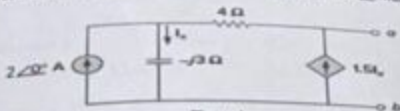


Figure 1

- The open circuit voltage between the a-b terminals is :
a. $(-24 + j12)$ V b. $(25 - j12)$ V c. $(12 + j14)$ V d. None of all
- The short circuit current (I_{sc}) when the a-b terminals are shorted is:
a. -20 A b. $(-2.5 + j2.6)$ A c. $(2.64 + j0.48)$ A d. None of all
- The Thevenin impedance (Z_{th}) between the a-b terminals is:
a. $(8 + j8)$ ohm b. $(-8 + j6)$ ohm c. $(3 + j2.5)$ ohm d. None of all
- The current through a load of $(6 - j4)$ ohm connected between the a - b terminals is:
a. $(9 + j3)$ A b. $(8 - j1)$ A c. $(15 + j4)$ A d. None of all

**Consider the circuit shown in Figure 2 where $V = 10\angle -60^\circ$ V rms, and $I_s = 5\angle 10^\circ$ A rms, and answer questions 5-7.

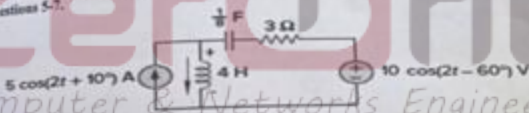


Figure 2

- The voltage across the inductor due to V source only (in V) is:
a. $12.9 - j8.29$ b. $14.71 - j6.29$ c. $12 + j5$ d. None of all
- The voltage across the inductor due to I_s source only (in V) is:
a. $-12.79 - j6.26$ b. 40 c. $39.76 - j4.36$ d. None of all
- The current through the inductor due to both sources in A is:
a. $8.71 + j6.29$ b. $14.71 - j6.29$ c. $15.4 - j3$ d. None of all

***Consider a 240 V rms, 60 Hz source supplies a load that is 10 kW(resistive), 15kVAR(capacitive) and 22kVAR (inductive) and answer questions 8-12.

- The power factor of the load is
a. 0.866 lagging b. 0.707 lagging c. 0.82 lagging d. None of all
- The current of the load is
a. $41.67 - j29.17$ A b. $34.75 - j12.0$ A c. $21 + j42.75$ A d. None of all
- The apparent power supplied by the source is
a. 32000 VA b. 12210 VA c. $1231.5 + j564.8$ VA d. None of all
- The capacitance required to improve the power factor to 0.96 lagging is
a. 188.93 μ F b. 393.2 μ F c. 24.87 μ F d. None of all

12. The current drawn from the source after the power factor correction is
 a. $41.3 - j10.75 \text{ A}$ b. $75 - j1.8 \text{ A}$ c. $41.67 - j12.15 \text{ A}$ d. None of all
- ** Consider a balanced three phase $\text{Y}-\Delta$ system has $V_{\text{rms}} = 120\angle 0^\circ \text{ V}_{\text{rms}}$ in positive sequence and a load of $Z_{\text{in}} = 81 + j45 \Omega$. If the line impedance is $0.4 + j1.35 \Omega$ as shown in Figure 3 and answer questions 13-17.

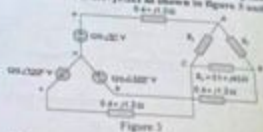


Figure 3

13. The voltage V_{in} is
 a. $173 - j216.5 \text{ V}$ b. $180 + j103.99 \text{ V}$ c. 240 V d. None of all
14. The line current I_{a} is
 a. $3.69 - j3.44 \text{ A}$ b. $4.25 + j4.33 \text{ A}$ c. $-2.65 - j3.5 \text{ A}$ d. None of all
15. The phase current I_{ab} is
 a. $3.54 + j0.54 \text{ A}$ b. $3.2 + j7.5 \text{ A}$ c. $2.84 - j0.63 \text{ A}$ d. None of all
16. The total average power supplied by the source is
 a. 4871 W b. 1329.5 W c. 8437 W d. None of all
17. The total complex power consumed by the load is
 a. $348 + j1.12 \text{ KVA}$ b. $1.3 + j1.47 \text{ KVA}$ c. $1.71 + j2.12 \text{ KVA}$ d. None of all

** Consider a series RLC circuit has $R = 4 \Omega$, $L = 25 \text{ mH}$, $C = 0.625 \mu\text{F}$, and $V_s = 15\angle 0^\circ \text{ V}_{\text{rms}}$ and answer questions 18-20 showing the required equations.

18. The resonant frequency (ω_0) in (rad/sec) and the quality factor (Q)
 19. The cutoff frequencies ω_1 and ω_2 (in rad/sec) are
 20. The voltage across the capacitor (V_C) at resonance (in V) is

** Consider the magnetically coupled circuit shown in Figure 4 and answer questions 21-25 showing the required equations.

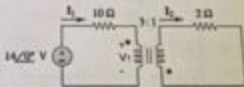


Figure 4

21. The equivalent input impedance (Z_{in}) seen by the source (in Ω)
 22. The current supplied by the source (I_1)
 23. The voltage (V_1) across the primary winding is:
 24. The power absorbed by the 2Ω resistor is:
 25. The complex power supplied by the source is:

1. a 2. c 3. b 4. a 5. b

6. c 7. d 8. c 9. Team 10. b

11. a 12. c 13. d 14. a 15. d

16. b 17. d

$$18. \omega_0 = \frac{1}{\sqrt{LC}} = 8000, \quad Q = \frac{1}{R} \sqrt{\frac{L}{C}} = 50$$

$$19. Q > 10 \checkmark \rightarrow \omega_1 = \omega_0 - \frac{B}{2}$$

$$\omega_2 = \omega_0 + \frac{B}{2}$$

$$B = \frac{R}{L} = \frac{\omega_0}{Q} = 160 \text{ Hz}$$

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$$\omega_1 = 7920$$

$$\omega_2 = 8080$$

Course Name: Electrical Circuits (2)
Instructors: Dr A. Alzyoud and Eng J. Abu-Ghalune

Date: Jan. 31, 2022
Exam Time: 90 minutes

Prob.. 1	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer													

Prob. 1: Choose the correct answer.

Consider the circuit shown in Figure 1, where the voltage source has $V_s = 5\angle 0^\circ \text{ V}_{rms}$.

- The open circuit voltage (V_o) between the a-b terminals is
a. $50(1-j1) \text{ V}$ b. $75(1-j1) \text{ V}$ c. $25(1-j1) \text{ V}$ d. None of all
- The short circuit current (I_{sc}) when the a-b terminals are shorted is
a. -20 A b. -10 A c. -30 A d. None of all
- The Thevenin impedance (Z_{th}) between the a-b terminals is
a. $2.5 + j2.5 \Omega$ b. $2.5 - j2.5 \Omega$ c. $-2.5 + j2.5 \Omega$ d. None of all
- The current through a load of $7.5 - j2.5 \Omega$ connected between the a - b terminals is
a. $5(1-j1) \text{ A}$ b. $10(1-j1) \text{ A}$ c. $15(1-j1) \text{ A}$ d. None of all

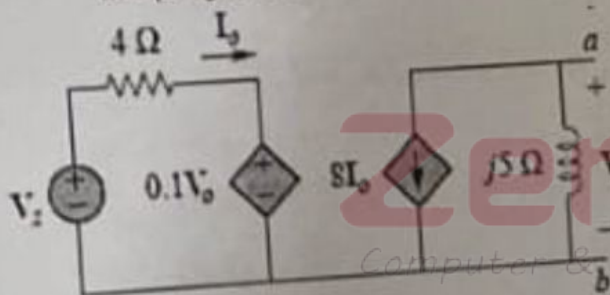


Figure 1

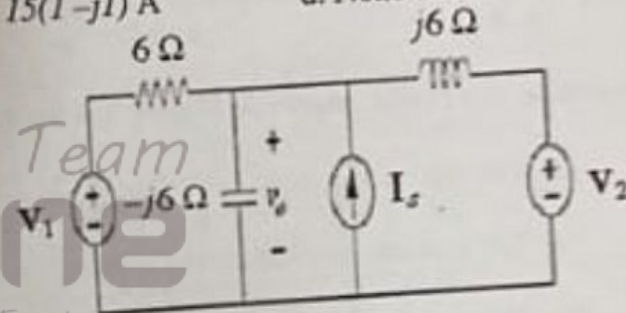


Figure 2

Consider the circuit shown in Figure 2 where $V_1 = 6\angle 0^\circ \text{ V}_{rms}$, $V_2 = 6\angle 0^\circ \text{ V}_{rms}$, and $I_s = 1\angle 0^\circ \text{ A}_{rms}$.

- The voltage across the capacitor due to V_1 source only (in V) is
a. 6 b. 3 c. 12 d. None of all
- The voltage across the capacitor due to V_2 source only (in V) is
a. $-j12$ b. $-j6$ c. $-j3$ d. None of all
- The voltage across the capacitor due to I_s source only (in V) is
a. 6 b. 12 c. 18 d. None of all

Three Loads connected in parallel across a voltage source of $200\angle 0^\circ \text{ V}_{rms}$, where the loads have:
Load 1: absorbs 250VA at 0.6 lagging p.f., Load 2: absorbs 200W at 0.8 leading p.f., and Load 3: absorbs 300VA and 150VAR.

- The power factor of the load 3 is
a. 0.866 lagging b. 0.707 lagging c. 0.5 lagging d. None of all
- The current of load 1 is
a. $1.3 - j0.75 \text{ A}$ b. $0.75 - j1.0 \text{ A}$ c. $1 + j0.75 \text{ A}$ d. None of all
- The complex power supplied by the source is
a. $564.8 + j501.2 \text{ VA}$ b. $609.8 + j200 \text{ VA}$ c. $231.5 + j564.8 \text{ VA}$ d. None of all
- The power factor of the equivalent load is
a. 0.95 lagging b. 0.748 lagging c. 0.38 lagging d. None of all
- The current supplied by the voltage source in rms is
a. $2.822 - j2.506 \text{ A}$ b. $2.822 - j1.157 \text{ A}$ c. $3.049 - j1.0 \text{ A}$ d. None of all
- The impedance of the equivalent load is
a. $24.87 - j60.67 \Omega$ b. $59.22 + j19.42 \Omega$ c. $60.67 + j24.87 \Omega$ d. None of all

Prob. 1	14	15	16	17	18	19	20	21	22	23	24	25	
Answer													

A balanced Δ -load of $Z_{\Delta} = 8.66 + j5 \Omega$ is supplied by a balanced Y voltages of $V_{an} = 250/\theta^\circ$ V_{rms} in positive sequence.

14. The voltage V_{AB} is
a. $375 + j216.5$ V b. $216.5 + j375$ V c. 433 V d. None of all
15. The phase current I_{BC} is
a. $37.5 + j21.65$ A b. $j43.3$ A c. $-21.65 - j37.5$ A d. None of all
16. The line current I_c is
a. $37.5 + j65$ A b. $j75$ A c. $37.5 - j65$ A d. None of all
17. The total average power supplied by the source is
a. 48.71 KW b. 28.21 KW c. 84.37 KW d. None of all
18. The total complex power consumed by the load is
a. $56.48 + j50.12$ KVA b. $56.48 + j23.15$ KVA c. $48.71 + j28.12$ KVA d. None of all

Consider a series RLC circuit has $R = 4 \Omega$, $L = 4$ mH, $C = 0.4 \mu$ F, and $V_s = 12/\theta^\circ$ V_{rms}.

19. The resonant frequency (ω_0) in (rad/sec) is
a. 25 K b. 50 K c. 20 K d. None of all
20. The quality factor (Q) is
a. 50 b. 30 c. 25 d. None of all
21. The cutoff frequencies ω_1 and ω_2 (in rad/sec) are
a. (19.2, 20.8) K b. (24, 26) K c. (49, 51) K d. None of all
22. The voltage across the capacitor (V_C) at resonance (in V) is
a. 400 b. 300 c. 12 d. None of all

Consider the circuit shown in Figure 3, where $V_s = 15/\theta^\circ$ V_{rms}, $R_1 = 2 \Omega$, and $R_L = 4 \Omega$.

23. The equivalent input impedance (Z_{in}) seen by the source (in Ω)
a. $4 + j3$ b. $3 + j3$ c. $5 + j3$ d. None of all
24. The current supplied by the source (I_1)
a. $2 - j1.2$ b. $1.6 - j1.2$ c. $2.5 - j2.5$ d. None of all
25. The power absorbed by the R_L resistor is
a. 12.5 W b. 10.88 W c. 8 W d. None of all

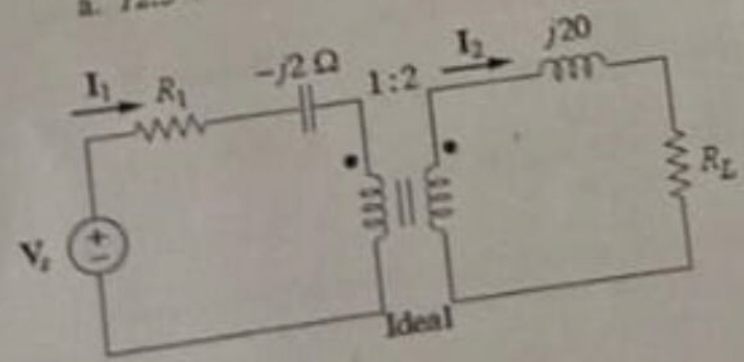


Figure 3

$$I_o = \frac{V_s - 0.1 V_o}{4} = \frac{5 - 0.1 V_o}{4} \quad \text{--- (1)}$$

$$V_o = -8 \bar{I}_o \times j5$$

$$= -40 j \bar{I}_o$$

$$\boxed{1-4}$$

$$V_o = -40 j \left(\frac{5 - 0.1 V_o}{4} \right)$$

$$V_o = 25 - j25 \quad \text{(c)}$$

$$\bar{I}_o = \frac{5 - 0.1 V_o}{4}$$

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$$I_{sc} = -8 \bar{I}_o = -8 \left(\frac{5 - 0.1 V_o}{4} \right) = -10 \text{ A} \quad \text{(b)}$$

$$Z_{th} = \frac{V_o}{\bar{I}_{sc}} = -2.5 + j2.5 \, \Omega \quad \text{(c)}$$

$$I = \frac{V_{th}}{Z_{th} + Z_L} = 5 - j5 \quad \text{(a)}$$

$$V_{c1} = V_1 = 6 \angle 0^\circ \quad (2)$$

$$V_{c2} = V_2 \times \frac{3-j3}{3+j3}$$

$$\boxed{5-j7}$$

$$= 6 \angle -90^\circ = -j6 \quad (3)$$

$$\frac{V_{c3}}{I_c} = I_s \times \frac{3-j3}{3+j3} = 1 \angle 90^\circ = -jA$$

$$V_{c3} = I_c \times X_c = -j \times -6j = 6V \quad (4)$$

Load 1:-

$$S = 250 \text{ VA}$$

0.6 lagging p.f

Load 2

$$P = 200 \text{ W}$$

0.8 leading p.f.

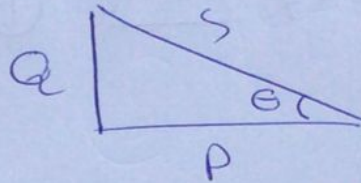
$$V_{\text{rms}} = 200 \angle 0^\circ$$

8-13

Load 3:-

$$S = 300 \text{ VA}$$

$$Q = 150 \text{ VAR}$$

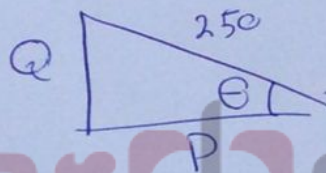


$$\sin \theta = \frac{Q}{S}$$

$$\theta = 30^\circ \quad \text{p.f.} = \cos(30^\circ) = 0.866 \quad (a)$$

$I_1 ??$

$$S = 250$$



$$\cos \theta = 0.6$$

$$\cos \theta = \frac{P}{S}$$

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$$P = 250 \times 0.6 = 150 \text{ W}$$

$$\tilde{S} = S \angle \theta = 250 \angle 53.13^\circ$$

$$\tilde{S} = \tilde{V}_{\text{rms}} \tilde{I}_{\text{rms}}^*$$

$$\tilde{I}_{\text{rms}} = \frac{\tilde{S}}{\tilde{V}_{\text{rms}}}^* = \frac{250 \angle -53.13^\circ}{200 \angle 0^\circ} = 0.75 - jA \quad (b)$$

$$\tilde{S} = \tilde{S}_1 + \tilde{S}_2 + \tilde{S}_3 = 250 + 250 + 300 \angle 18.16^\circ$$

$$= 641.8 \angle 18.16^\circ \quad (b)$$

$$\cos(18.16) = 0.95 \text{ lagging} \quad (a)$$

$$I_{rms} = \frac{S_{tot}}{V_{rms}} = 3.209 \angle -18.16^\circ \quad (c)$$

$$Z = \frac{V}{I} = 62.3247 \angle 18.16 \quad (b)$$

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$$Z_{\Delta} = 8.66 + j5 \Omega$$

P.D.S.

14-18

$$V_{an} = 250 \angle 0^\circ V_{rms}$$

$$\begin{aligned} V_{ab} = V_{AB} &= \sqrt{3} V_{an} \angle 30^\circ \\ &= (\sqrt{3} \angle 30^\circ) \times (250 \angle 0^\circ) \\ &= 375 + j216.5 \text{ v. } (a) \end{aligned}$$

$I_{ac}??$

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}} = \frac{375 + j216.5}{8.66 + j5} = 43 \angle -7 \times 10^{-4} \text{ A}$$

$$I_{CA} = I_{AB} \angle 120^\circ = -21.65 + j37.5 \text{ A} \quad (b)$$

$$I_{AC} = -21.65 - j37.5 \text{ A} \checkmark$$

$$I_c = I_{cn} = I_{an} \angle 120^\circ$$

$$I_{cn} = \frac{V_{an}}{Z_{\Delta/3}} \angle 120^\circ = \frac{250 \angle 120^\circ}{2.88 + j1.66} \approx j75 \text{ A} \quad (b)$$

$$\begin{aligned} P &= 3 I^2 Z_{\Delta} = 3 \times 43.33^2 \times 8.66 \\ &= 48.71 \text{ K watt.} \end{aligned}$$

$$\tilde{S} = \frac{P}{\cos \theta} = 48.71 + j28.12 \text{ KVA} \quad (c)$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4m \times 0.4\mu}} = 25000 = 25k \text{ (a)}$$

$$Q = \frac{\omega_0 L}{R} = \frac{25k \times 4 \times 10^{-3}}{4} = 25 \text{ (c)} \quad Q \gg 10 \checkmark$$

$$\omega_1 = \omega_0 - \frac{B}{2}, \quad \omega_2 = \omega_0 + \frac{B}{2}$$

$$B = 1000 = \frac{25000}{25} = \frac{\omega_0}{Q}$$

19-22

$$\omega_1 = 25k - 500 = 24.5k \text{ (d)}$$

$$\omega_2 = 25k + 500 = 25.5k$$

$$V_c = V_m Q = 12 \angle 0^\circ \times 25 = 300 \text{ (b)}$$

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$$Z_{in} = Z_R + R_1 + j\omega L_1$$

$$Z_R = \frac{Z_L}{n^2}$$

$$= 1 + j5 + 2 - j2$$

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$$= \frac{j20 + 4}{4} = \underline{\underline{1 + j5}}$$

$$= 3 + j3 \text{ (b)}$$

$$I_1 = \frac{V}{Z_{in}} = \frac{15 \angle 0}{3 + j3} = 2.5 - j2.5 \text{ (c)}$$

23-25

$$P = V_o I_2$$

$$I_2 = +\frac{I_1}{n} = +1.25 + j1.25$$

$$V_o = I_2 \times R_L = +5 + j5$$

$$P = I V = 12.5 \text{ W (a)}$$